Hanpeng Liu

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Research Summary

My research focuses on visual representation learning for multimodal large language models (MLLMs), spanning image-text pretraining [C1], CLIP post-training [C2], self-supervised visual pretraining [J1, P1], and instruction-aware vision encoding for multimodal understanding [P2]. My publications include three NeurIPS papers and one CVPR paper; I am first or co-first author on four papers and corresponding author on one. Going forward, I aim to extend this line of work toward multimodal understanding or image generation.

Interests: Vision encoders for multimodal LLMs · Vision-language pretraining (CLIP) · Self-supervised visual representation learning · Efficient architectures and knowledge distillation

Education

Huazhong University of Science and Technology (HUST) Sep 2022 – Jun 2027 (expected)
Ph.D. in Computer Science and Technology (direct-entry program); Advisor: Prof. Kun He *Wuhan, China*

Huazhong University of Science and Technology (HUST) Sep 2018 – Jun 2022
B.Eng. in Computer Science and Technology; Rank: 23/373; Outstanding Graduate *Wuhan, China*

Publications

[*] Equal contribution [†] Corresponding author

Conference Papers

[C1] **Hanpeng Liu**, Yaqian Li[†], Zidan Wang, Shuoxi Zhang, Zonglin Zhao, Zihao Bo, Rinyoichi Takezoe, Kaiwen Long, Kun He[†]. *ITO: Multi-View Alignment and Training-Time Fusion for Image-Text Pretraining*. Advances in Neural Information Processing Systems (**NeurIPS**), 2026. [Paper] [Code] [Project]

[C2] Zidan Wang, Yaqian Li[†], Xiaokai Zhang, Kaiwen Long, Kun He, **Hanpeng Liu**[†]. *Rethinking Contrastive Loss in CLIP Post-training: A Complementary Framework with Frozen Text Encoder*. Advances in Neural Information Processing Systems (**NeurIPS**), 2026. [Paper] [Code] [Project]

[C3] Shuoxi Zhang, **Hanpeng Liu**, Stephen Lin, Kun He[†]. *Hyperspherical Dataset Distillation via Contrastive Embedding Alignment*. IEEE International Conference on Multimedia and Expo (**ICME**), 2025. [Paper]

[C4] Jinsong Chen, **Hanpeng Liu**, John E. Hopcroft, Kun He. *Leveraging Contrastive Learning for Enhanced Node Representations in Tokenized Graph Transformers*. Advances in Neural Information Processing Systems (**NeurIPS**), 2024. [Paper]

[C5] Shuoxi Zhang, **Hanpeng Liu**, Yuyi Wang, Kun He[†]. *Grouped Logit Distillation Enhanced with Superclass Awareness for Efficient Knowledge Transfer*. European Conference on Artificial Intelligence (**ECAI**), 2024. [Paper]

[C6] Shuoxi Zhang[*], **Hanpeng Liu**[*], Stephen Lin, Kun He. *You Only Need Less Attention at Each Stage in Vision Transformers*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024. [Paper]

Journal Papers

[J1] **Hanpeng Liu**, Shuoxi Zhang, Kaiyuan Gao, Kun He[†]. *Boosting the Patch-based Self-Supervised Learning through Past-to-Present Smoothing*. **Pattern Recognition**, 2025.

[J2] Shuoxi Zhang, **Hanpeng Liu**, Yuyi Wang, Kun He, Jun Lin, Yang Zeng. *Class Discriminative Knowledge Distillation*. **IEEE Transactions on Emerging Topics in Computational Intelligence**, 9(2):1340–1351, 2025. [Paper]

[J3] **Hanpeng Liu**[*], Shuoxi Zhang[*], Kun He[†]. *Residual Learning for Self-Knowledge Distillation*. **IEEE Transactions on Big Data**, 2025.

[J4] Shuoxi Zhang, **Hanpeng Liu**, Kun He. *Knowledge Distillation via Token-Level Relationship Graph Based on the Big Data Technologies*. **Big Data Research**, 2024.

Preprints

[P1] **Hanpeng Liu**, Zidan Wang, Shuoxi Zhang, Kaiyuan Gao, Kun He. *Separators in Enhancing Autoregressive Pretraining for Vision Mamba*. arXiv:2603.03806, 2026. [Paper]

[P2] **Hanpeng Liu**, Yaqian Li†, Zidan Wang, Shuoxi Zhang, Zihao Bo, Rinyoichi Takezoe, Kaiwen Long, Kun He†. *iGVLM: Dynamic Instruction-Guided Vision Encoding for Question-Aware Multimodal Understanding*. arXiv:2603.02748, 2026. [Paper]

Research Experience

Li Auto Inc. Sep 2024 – Present
Research Intern *China*

- Contributed to building a distributed training and evaluation platform for vision and vision-language foundation models (DDP / FSDP / DeepSpeed), with standardized benchmarks on ImageNet and COCO.
- Led the scaling of CLIP-style image-text pretraining to **128 NVIDIA A800 GPUs** on **1B-scale** image-text data, using gradient-preserving all-gather for the contrastive loss and WebDataset streaming pipelines.

Mentoring

- **CLIP post-training** [C2] (NeurIPS 2026, co-corresponding author): proposed the research idea, mentored junior HUST students through the project, contributed to the experiments, and led the paper writing.

Academic Service

- **Conference Reviewer:** NeurIPS, ICLR, CVPR, AAAI
- **Journal Reviewer:** Pattern Recognition, IEEE Transactions on Cybernetics

Honors and Awards

Merit Graduate Student & Zhixing Scholarship, HUST	2023, 2024
Outstanding Graduate, HUST	2022
Third Prize (National), China Post-Graduate Mathematical Contest in Modeling	2022
Third Prize (National), Bisheng Cup Compiler System Design Competition (Team Leader)	2021
First Prize (National), Chinese Mathematics Competitions	2020

Skills

Programming: Python, PyTorch, timm, Hugging Face Transformers

Large-scale training: Multi-GPU / multi-node distributed training (DDP, FSDP, DeepSpeed ZeRO), mixed-precision training, WebDataset streaming pipelines; LLaVA-style MLLM training with Qwen2 backbones

References

Prof. Kun He
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Executive Director, Hopcroft Center on Computing Science, HUST
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Ph.D. advisor

Additional references available upon request.